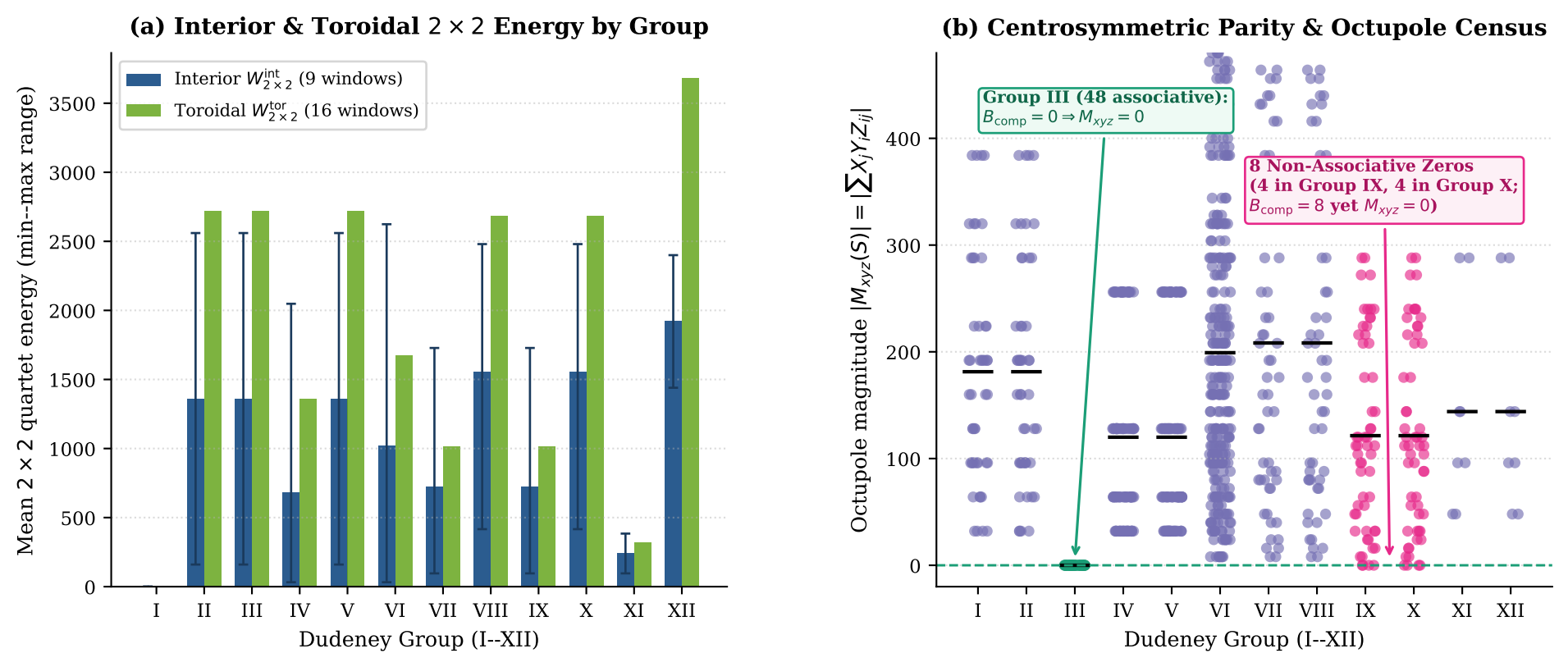

(a) Interior & Toroidal 2 × 2 Energy by Group
Interior $W_{2\times2}^{\mathrm{int}}$ (9 windows)
Toroidal $W_{2\times2}^{\mathrm{tor}}$ (16 windows)
Mean 2 × 2 quartet energy (min--max range)
0
500
1000
1500
2000
2500
3000
3500
I
II
III
IV
V
VI
VII
VIII
IX
X
XI
XII
Dudeney Group (I--XII)
(b) Centrosymmetric Parity & Octupole Census
Group III (48 associative):
$B_{\mathrm{comp}} = 0 \Rightarrow M_{xyz} = 0$
8 Non-Associative Zeros
(4 in Group IX, 4 in Group X;
$B_{\mathrm{comp}} = 8$ yet $M_{xyz} = 0$)
Octupole magnitude $|M_{xyz}(S)| = |\sum X_i Y_i Z_i|$
0
100
200
300
400
I
II
III
IV
V
VI
VII
VIII
IX
X
XI
XII
Dudeney Group (I--XII)